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FINAL EXAMINATION
SEMESTER 2 SESSION 2016/2017

CODE: MES3103 **COURSE: SOFTWARE VERIFICATION AND VALIDATION**

DATE: 07 JUN 2017 **DURATION: 2 HOURS**

INSTRUCTIONS

1. This exam paper consists of only **ONE (1)** part only, namely **Part A**.
2. Answer **ALL** questions in the answer booklet provided.
3. This question paper will be collected at the end of the examination.

This question paper consists of 2 pages including cover

PROGRAMME: _____

YEAR: _____ LECTURE GROUP: A

REGISTRATION NO.: _____ [ ][ ][ ][ ][ ][ ]

IDENTITY CARD NO.: [ ][ ][ ][ ][ ][ ] [ ][ ][ ][ ][ ][ ]

LECTURER: DR. AZNIAH BINTI ISMAIL

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PART A: (40 marks)

Instruction: Answer **all** questions in the answer booklet provided.

1. Explain the differences between *verification* and *validation.*

[5 marks]

2. Explain the differences between *debug* and *failure.*

[5 marks]

3. The modern verification has four major categories of fundamental techniques.

   (a) List all the **four (4)** categories.

   [4 marks]

   (b) Briefly describe **two (2)** of the categories.

   [6 marks]

4. Given the requirements and functional specifications in *Figure 1*, suggest any **four (4)** different ways of how we can verify and validate any of them.

   (Make sure: (i) you clearly state whether each of your suggestion is under verification or validation, and (ii) there must be a mixture of verification and validation in the whole suggestion. No long explanation is required so you may use a table to represent your answer.)

   [10 marks]

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Requirement specification *User wants to control the lights in 4 rooms by remote command sent from the UI for each room separately.*

Functional specification
RUI 01.The UI will contain 4 checkboxes labelled according to rooms they control.

RUI 02. When a checkbox is checked, the signal is sent to corresponding light. A green dot appears next to the checkbox

RUI 03. When a checkbox is unchecked, the signal (turn off) is sent to corresponding light. A red dot appears next to the checkbox.

Figure 1 – Requirements and functional specifications for light controller UI.

5. There are many different types of testing for validation purposes. Choose only **two (2)** of them and make comparisons to show their differences.

   [10 marks]

END OF QUESTIONS

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